

Mean

A **mean** is a quantity representing the "center" of a collection of numbers and is intermediate to the extreme values of the set of numbers.^[1] There are several kinds of **means** (or "measures of central tendency") in mathematics, especially in statistics. Each attempts to summarize or typify a given group of data, illustrating the magnitude and sign of the data set. Which of these measures is most illuminating depends on what is being measured, and on context and purpose.^[2]

The arithmetic mean, also known as "arithmetic average", is the sum of the values divided by the number of values. The arithmetic mean of a set of numbers $x_1, x_2, \dots, x_n$ is typically denoted using an overhead bar, $\bar{x}$.^[note 1] If the numbers are from observing a sample of a larger group, the arithmetic mean is termed the sample mean ($\bar{x}$) to distinguish it from the group mean (or expected value) of the underlying distribution, denoted μ or μ_x .^{[note 2][3]}

Outside probability and statistics, a wide range of other notions of mean are often used in geometry and mathematical analysis; examples are given below.

Types of means

Pythagorean means

In mathematics, the three classical **Pythagorean means** are the arithmetic mean (AM), the geometric mean (GM), and the harmonic mean (HM). These means were studied with proportions by Pythagoreans and later generations of Greek mathematicians^[4] because of their importance in geometry and music.

Arithmetic mean (AM)

The arithmetic mean (or simply *mean* or *average*) of a list of numbers, is the sum of all of the numbers divided by their count. Similarly, the mean of a sample $x_1, x_2, \dots, x_n$, usually denoted by $\bar{x}$, is the sum of the sampled values divided by the number of items in the sample.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{x_1 + x_2 + \dots + x_n}{n}$$

For example, the arithmetic mean of five values: 4, 36, 45, 50, 75 is:

$$\frac{4 + 36 + 45 + 50 + 75}{5} = \frac{210}{5} = 42.$$

Geometric mean (GM)

The geometric mean is an average that is useful for sets of positive numbers, that are interpreted according to their product (as is the case with rates of growth) and not their sum (as is the case with the arithmetic mean).^[1]

$$\bar{x} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}} = (x_1 x_2 \cdots x_n)^{\frac{1}{n}}$$

For example, the geometric mean of five values: 4, 36, 45, 50, 75 is:

$$(4 \times 36 \times 45 \times 50 \times 75)^{\frac{1}{5}} = \sqrt[5]{24\,300\,000} = 30.$$

Harmonic mean (HM)

The harmonic mean is an average which is useful for sets of numbers which are defined in relation to some unit, as in the case of speed (i.e., distance per unit of time):

$$\bar{x} = n \left(\sum_{i=1}^n \frac{1}{x_i} \right)^{-1}$$

For example, the harmonic mean of the five values: 4, 36, 45, 50, 75 is

$$\frac{5}{\frac{1}{4} + \frac{1}{36} + \frac{1}{45} + \frac{1}{50} + \frac{1}{75}} = \frac{5}{\frac{1}{3}} = 15.$$

If we have five pumps that can empty a tank of a certain size in respectively 4, 36, 45, 50, and 75 minutes, then the harmonic mean of **15** tells us that these five different pumps working together will pump at the same rate as five pumps that can each empty the tank in **15** minutes.

Relationship between AM, GM, and HM

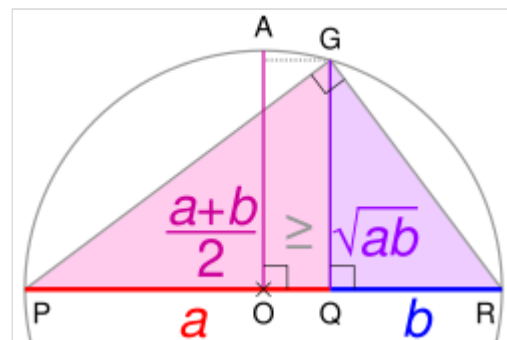
AM, GM, and HM of nonnegative real numbers satisfy these inequalities:^[5]

$$AM \geq GM \geq HM$$

Equality holds if all the elements of the given sample are equal.

Statistical location

In descriptive statistics, the mean may be confused with the median, mode or mid-range, as any of these may colloquially be called an "average" (more formally, a measure of central tendency). The mean of a set of observations is the arithmetic average of the values; however, for skewed distributions, the mean is not necessarily the same as the middle value (median), or the most likely value (mode). For example, mean income is typically skewed upwards by a small number of people with very large incomes, so that the majority have an income lower than the mean. By contrast, the median income is the level at which half the population is below and half is above. The mode income is the most likely income and favors the larger number of



Proof without words of the AM–GM inequality: □

PR is the diameter of a circle centered on O; its radius AO is the arithmetic mean of a and b. Triangle PGR is a right triangle from Thales's theorem, enabling use of the geometric mean theorem to show that its altitude GQ is the geometric mean. For any ratio a:b, AO ≥ GQ.

people with lower incomes. While the median and mode are often more intuitive measures for such skewed data, many skewed distributions are in fact best described by their mean, including the exponential and Poisson distributions.

Mean of a probability distribution

The mean of a probability distribution is the long-run arithmetic average value of a random variable having that distribution. If the random variable is denoted by $\mathbf{X}$, then the mean is also known as the expected value of $\mathbf{X}$ (denoted $E(\mathbf{X})$). For a discrete probability distribution, the mean is given by $\sum \mathbf{x}P(\mathbf{x})$, where the sum is taken over all possible values of the random variable and $P(\mathbf{x})$ is the probability mass function. For a continuous distribution, the mean is $\int_{-\infty}^{\infty} \mathbf{x}f(\mathbf{x})d\mathbf{x}$, where $f(\mathbf{x})$ is the probability density function.^[7] In all cases, including those in which the distribution is neither discrete nor continuous, the mean is the Lebesgue integral of the random variable with respect to its probability measure. The mean need not exist or be finite; for some probability distributions the mean is infinite ($+\infty$ or $-\infty$), while for others the mean is undefined.

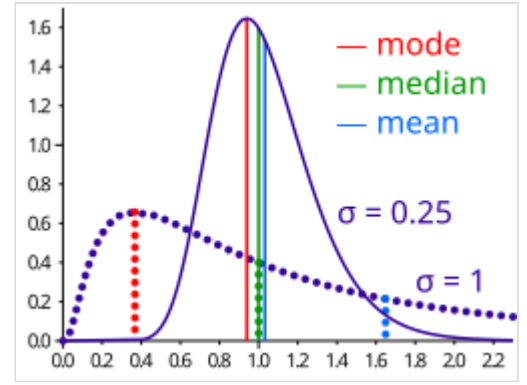
Generalized means

Power mean

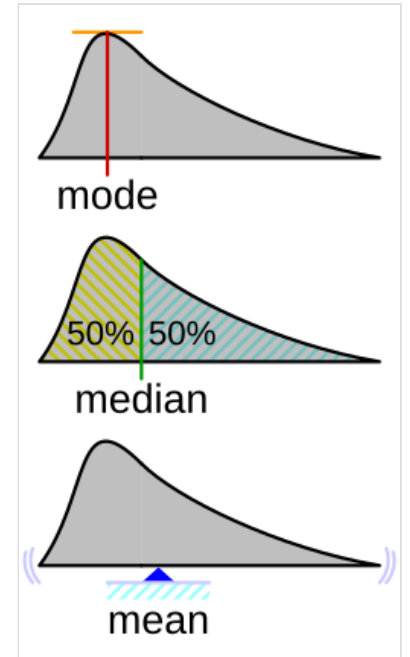
The generalized mean, also known as the power mean or Hölder mean, abstracts several other means. It is defined for positive numbers $x_1, \dots, x_n$ by^[1]

$$M_p(x_1, \dots, x_n) = \left(\frac{1}{n} \sum_{i=1}^n x_i^p \right)^{1/p}.$$

This, as a function of p , is well defined on $\mathbb{R} \setminus \{0\}$, but can be extended continuously to $\mathbb{R} \cup \{-\infty, +\infty\}$.^[8] By choosing different values for p , other well known means are retrieved.



Comparison of the arithmetic mean, median, and mode of two skewed (log-normal) distributions



Geometric visualization of the mode, median and mean of an arbitrary probability density function^[6]

Name	Exponent	Value
<u>Minimum</u>	$p = -\infty$	$\min\{x_1, \dots, x_n\}$
<u>Harmonic mean</u>	$p = -1$	$\frac{n}{\frac{1}{x_1} + \dots + \frac{1}{x_n}}$
<u>Geometric mean</u>	$p = 0$	$\sqrt[n]{x_1 \dots x_n}$
<u>Arithmetic mean</u>	$p = 1$	$\frac{x_1 + \dots + x_n}{n}$
<u>Root mean square</u>	$p = 2$	$\sqrt{\frac{x_1^2 + \dots + x_n^2}{n}}$
<u>Cubic mean</u>	$p = 3$	$\sqrt[3]{\frac{x_1^3 + \dots + x_n^3}{n}}$
<u>Maximum</u>	$p = +\infty$	$\max\{x_1, \dots, x_n\}$

Quasi-arithmetic mean

A similar approach to the power mean is the f -mean, also known as the quasi-arithmetic mean. For an injective function $f: I \rightarrow \mathbb{R}$ on an interval $I \subset \mathbb{R}$ and real numbers $x_1, \dots, x_n \in I$ we define their f -mean as

$$M_f(x_1, \dots, x_n) = f^{-1} \left(\frac{1}{n} \sum_{i=1}^n f(x_i) \right).$$

By choosing different functions f , other well known means are retrieved.

Mean	I	Function ^[note 3]
<u>Arithmetic mean</u>	$\mathbb{R}$	$x \mapsto x$
<u>Geometric mean</u>	$]0, +\infty[^{[note 4]}$	$x \mapsto \ln(x)$
<u>Harmonic mean</u>	$\mathbb{R} \setminus \{0\}$	$x \mapsto x^{-1}$
<u>Power mean</u>	$\mathbb{R} \setminus \{0\}^{[note 5]}$	$x \mapsto x^m$

Weighted arithmetic mean

The weighted arithmetic mean (or weighted average) is used if one wants to combine average values from different sized samples of the same population, and is define by^[1]

$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i},$$

where x_i and w_i are the mean and size of sample i respectively. In other applications, they represent a measure for the reliability of the influence upon the mean by the respective values.

Truncated mean

Sometimes, a set of numbers might contain outliers. Often, outliers are erroneous data caused by artifacts. In this case, one can use a truncated mean. It involves discarding given parts of the data at the top or the bottom end, typically an equal amount at each end and then taking the arithmetic mean of the remaining data. A specific example of a truncated mean is the interquartile mean.

Mean of a function

In some circumstances, mathematicians may calculate a mean of an infinite (or even an uncountable) set of values. This can happen when calculating the mean value y_{avg} of a function $f(x)$. Intuitively, a mean of a function can be thought of as calculating the area under a section of a curve, and then dividing by the length of that section. This can be done crudely by counting squares on graph paper, or more precisely by integration. The integration formula is written as:

$$y_{\text{avg}}(a, b) = \frac{1}{b - a} \int_a^b f(x) dx.$$

In this case, care must be taken to make sure that the integral converges. But the mean may be finite even if the function itself tends to infinity at some points.

Mean of angles and cyclical quantities

Angles, times of day, and other cyclical quantities require modular arithmetic to add and otherwise combine numbers. These quantities can be averaged using the circular mean. In all these situations, it is possible that no mean exists, for example if all points being averaged are equidistant. Consider a color wheel—there is no mean to the set of all colors. Additionally, there may not be a *unique* mean for a set of values: for example, when averaging points on a clock, the mean of the locations of 11:00 and 13:00 is 12:00, but this location is equivalent to that of 00:00.

Fréchet mean

The Fréchet mean gives a manner for determining the "center" of a mass distribution on a surface or, more generally, Riemannian manifold. Unlike many other means, the Fréchet mean is defined on a space whose elements cannot necessarily be added together or multiplied by scalars.

Center of a triangle

In geometry, there are thousands of different definitions for the center of a triangle that can all be interpreted as the mean of a triangular set of points in the plane.^[9]

Swanson's rule

This is an approximation to the mean for a moderately skewed distribution.^[10] It is used in hydrocarbon exploration and is defined as:

$$m = 0.3P_{10} + 0.4P_{50} + 0.3P_{90}$$

where P_{10} , P_{50} and P_{90} are the 10th, 50th and 90th percentiles of the distribution, respectively.

Other means

- [Arithmetic-geometric mean](#)
- [Arithmetic-harmonic mean](#)
- [Cesàro mean](#)
- [Chisini mean](#)
- [Contraharmonic mean](#)
- [Elementary symmetric mean](#)
- [Geometric-harmonic mean](#)
- [Grand mean](#)
- [Heinz mean](#)
- [Heronian mean](#)
- [Identric mean](#)
- [Lehmer mean](#)
- [Logarithmic mean](#)
- [Moving average](#)
- [Neuman–Sándor mean](#)
- [Quasi-arithmetic mean](#)
- [Root mean square \(quadratic mean\)](#)
- [Rényi's entropy \(a generalized f-mean\)](#)
- [Spherical mean](#)
- [Stolarsky mean](#)
- [Weighted geometric mean](#)
- [Weighted harmonic mean](#)

See also



- [Statistical dispersion](#)
- [Central tendency](#)
 - [Median](#)
 - [Mode](#)
- [Descriptive statistics](#)
- [Kurtosis](#)
- [Law of averages](#)
- [Mean value theorem](#)
- [Moment \(mathematics\)](#)
- [Summary statistics](#)
- [Taylor's law](#)

Notes

1. Pronounced "x bar".
2. Greek letter μ , pronounced /'mju:/.
3. For this column we will use the *mapping arrow* to denote a function. Under this notation, the function f is denoted by $x \mapsto f(x)$.
4. The geometric mean is well defined on $[0, +\infty[$, but this is not captured by this approach.
5. For $m \neq 0$ the domain can be $\mathbb{R}$.

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